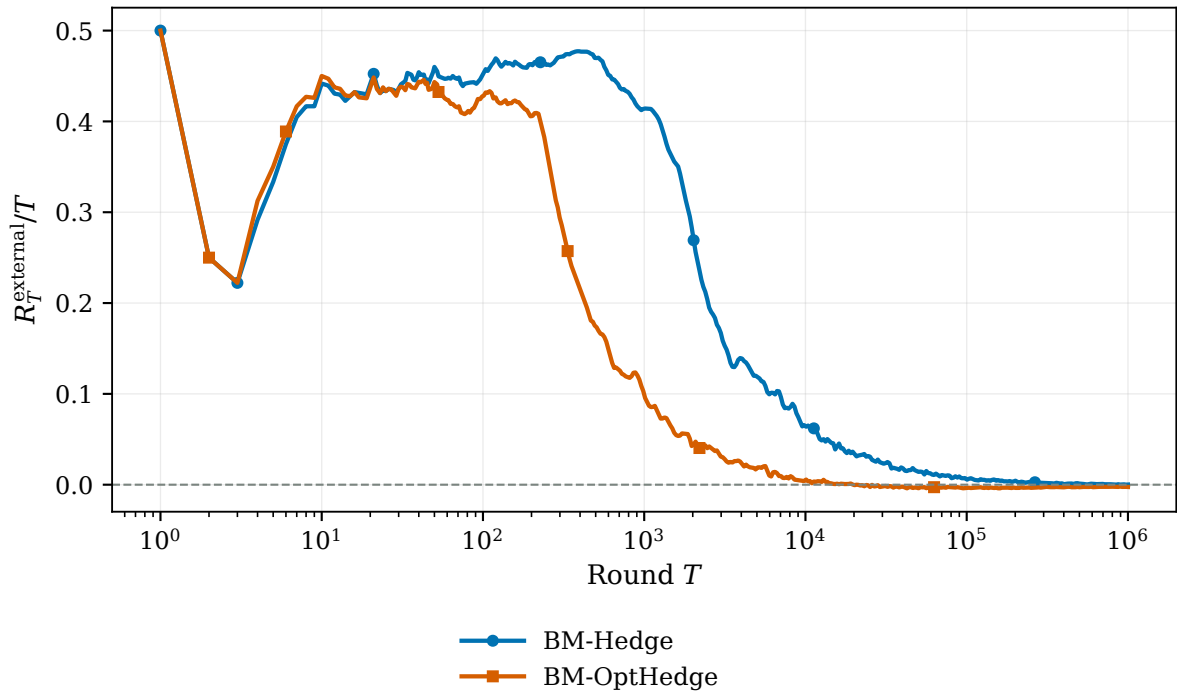


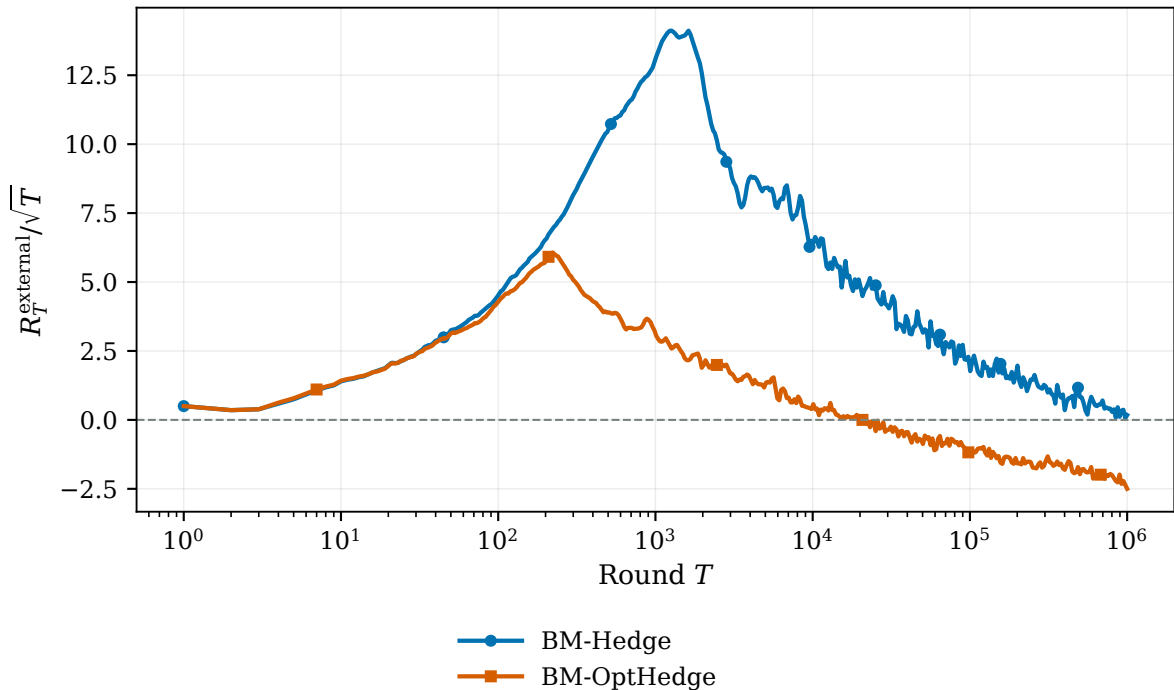
Experiment Information

Figure:	Algorithm comparison
Environment:	Historical-frequency
Actions:	10
Feedback:	Full information
Algorithms:	BM-Hedge, BM-OptHedge
Horizon:	1,000,000
Replicates:	12
Seed:	42
Metric:	External regret
View:	R / T
Final endpoints at T:	1,000,000
Endpoint statistics:	replicate mean, sample SD (ddof=1), SE=SD/sqrt(n)
BM-Hedge:	mean = 0.000175083, SD = 0.000465202, SE = 0.000134292
BM-OptHedge:	mean = -0.00249917, SD = 0.00047666, SE = 0.0001376



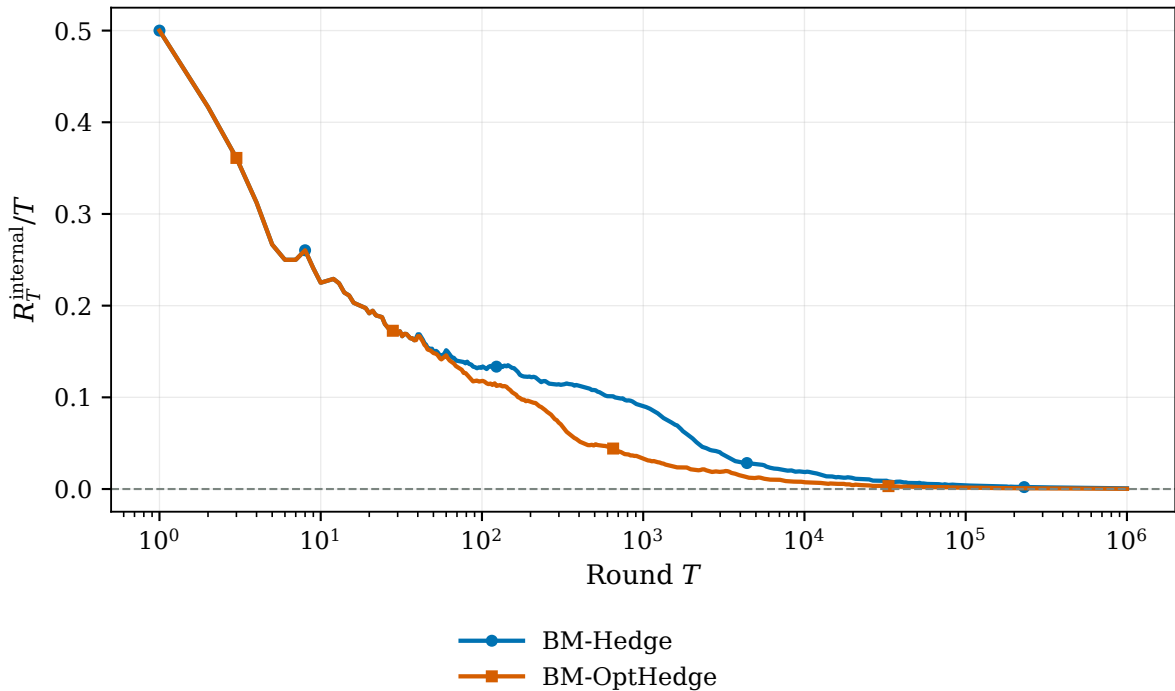
Experiment Information

Figure:	Algorithm comparison
Environment:	Historical-frequency
Actions:	10
Feedback:	Full information
Algorithms:	BM-Hedge, BM-OptHedge
Horizon:	1,000,000
Replicates:	12
Seed:	42
Metric:	External regret
View:	$R / \sqrt{T}$
Final endpoints at T:	1,000,000
Endpoint statistics:	replicate mean, sample SD (ddof=1), SE=SD/sqrt(n)
BM-Hedge:	mean = 0.175083, SD = 0.465202, SE = 0.134292
BM-OptHedge:	mean = -2.49917, SD = 0.47666, SE = 0.1376



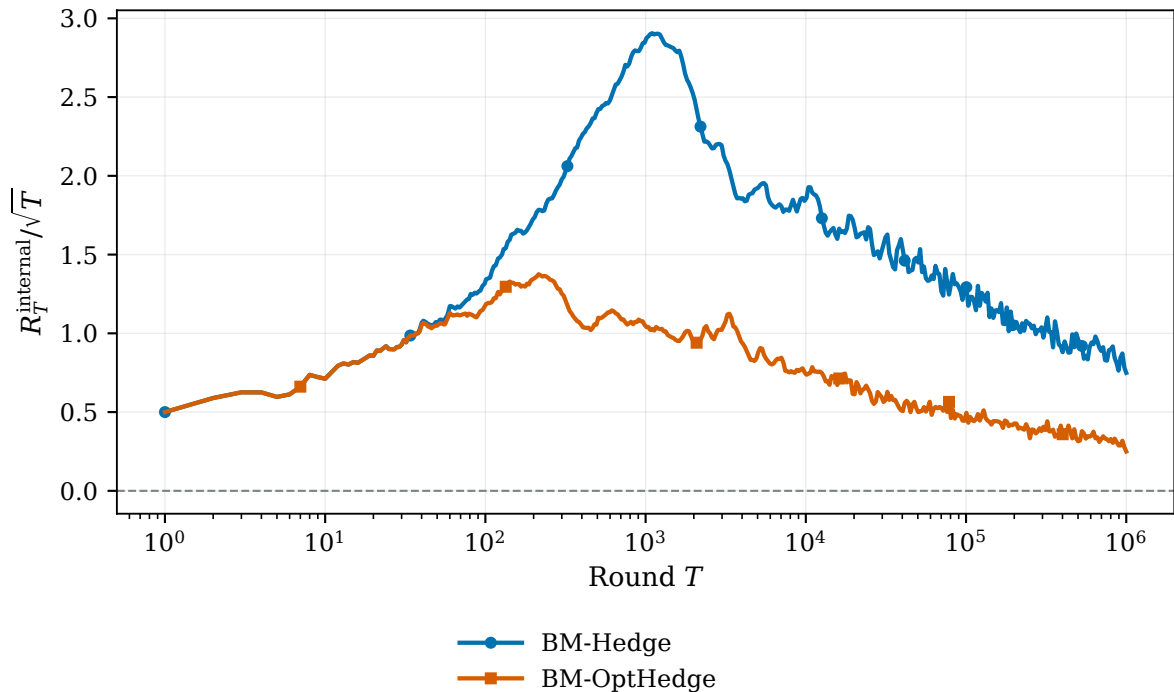
Experiment Information

Figure:	Algorithm comparison
Environment:	Historical-frequency
Actions:	10
Feedback:	Full information
Algorithms:	BM-Hedge, BM-OptHedge
Horizon:	1,000,000
Replicates:	12
Seed:	42
Metric:	Internal regret
View:	R / T
Final endpoints at T:	1,000,000
Endpoint statistics:	replicate mean, sample SD (ddof=1), SE=SD/sqrt(n)
BM-Hedge:	mean = 0.000749333, SD = 0.000111192, SE = 3.20984e-05
BM-OptHedge:	mean = 0.000249833, SD = 8.82711e-05, SE = 2.54817e-05



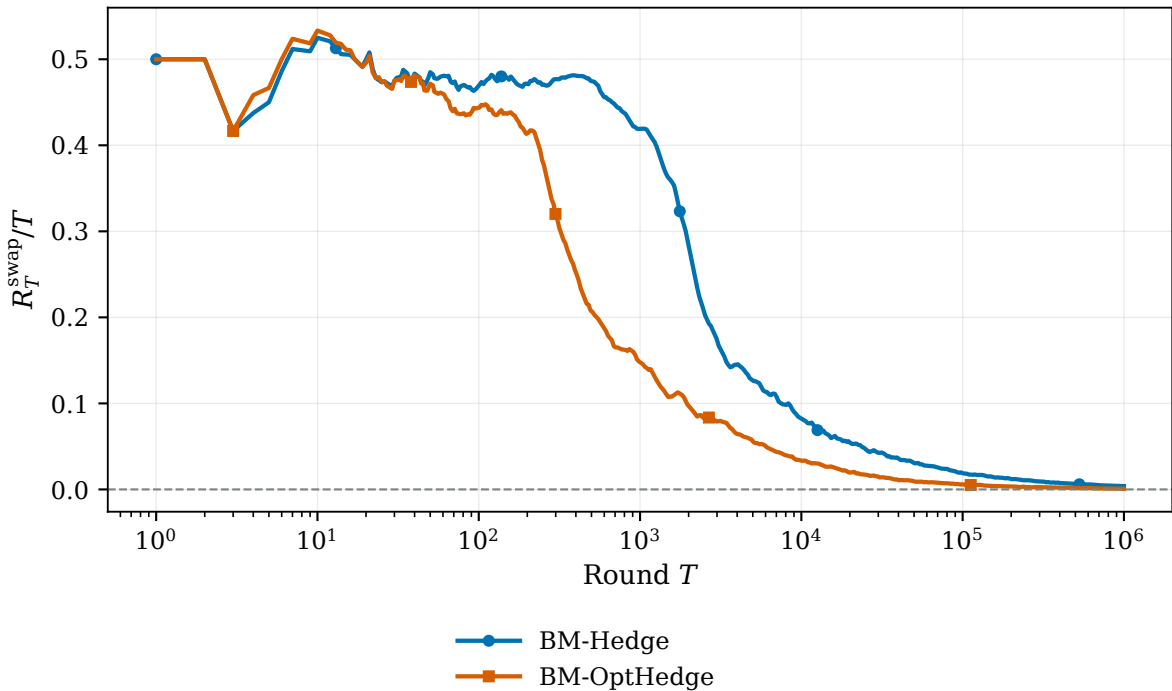
Experiment Information

Figure:	Algorithm comparison
Environment:	Historical-frequency
Actions:	10
Feedback:	Full information
Algorithms:	BM-Hedge, BM-OptHedge
Horizon:	1,000,000
Replicates:	12
Seed:	42
Metric:	Internal regret
View:	$R / \sqrt{T}$
Final endpoints at T:	1,000,000
Endpoint statistics:	replicate mean, sample SD (ddof=1), SE=SD/sqrt(n)
BM-Hedge:	mean = 0.749333, SD = 0.111192, SE = 0.0320984
BM-OptHedge:	mean = 0.249833, SD = 0.0882711, SE = 0.0254817



Experiment Information

Figure:	Algorithm comparison
Environment:	Historical-frequency
Actions:	10
Feedback:	Full information
Algorithms:	BM-Hedge, BM-OptHedge
Horizon:	1,000,000
Replicates:	12
Seed:	42
Metric:	Swap regret
View:	R / T
Final endpoints at T:	1,000,000
Endpoint statistics:	replicate mean, sample SD (ddof=1), SE=SD/sqrt(n)
BM-Hedge:	mean = 0.00389417, SD = 0.000468079, SE = 0.000135123
BM-OptHedge:	mean = 0.000729333, SD = 0.000297851, SE = 8.59821e-05



Experiment Information

Figure:	Algorithm comparison
Environment:	Historical-frequency
Actions:	10
Feedback:	Full information
Algorithms:	BM-Hedge, BM-OptHedge
Horizon:	1,000,000
Replicates:	12
Seed:	42
Metric:	Swap regret
View:	$R / \sqrt{T}$
Final endpoints at T:	1,000,000
Endpoint statistics:	replicate mean, sample SD (ddof=1), SE=SD/sqrt(n)
BM-Hedge:	mean = 3.89417, SD = 0.468079, SE = 0.135123
BM-OptHedge:	mean = 0.729333, SD = 0.297851, SE = 0.0859821

